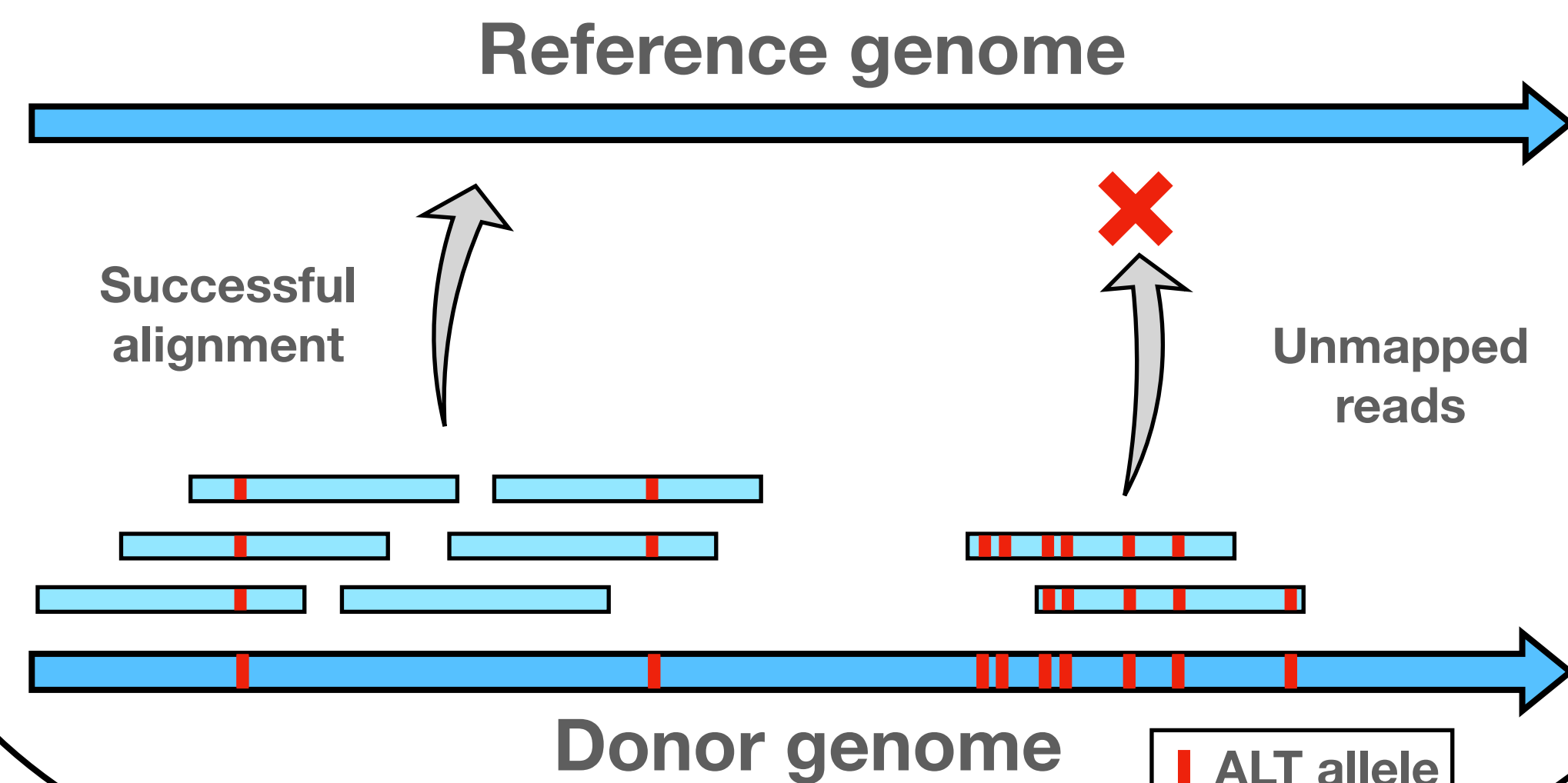


0 Reference bias



Measuring & Diagnosing

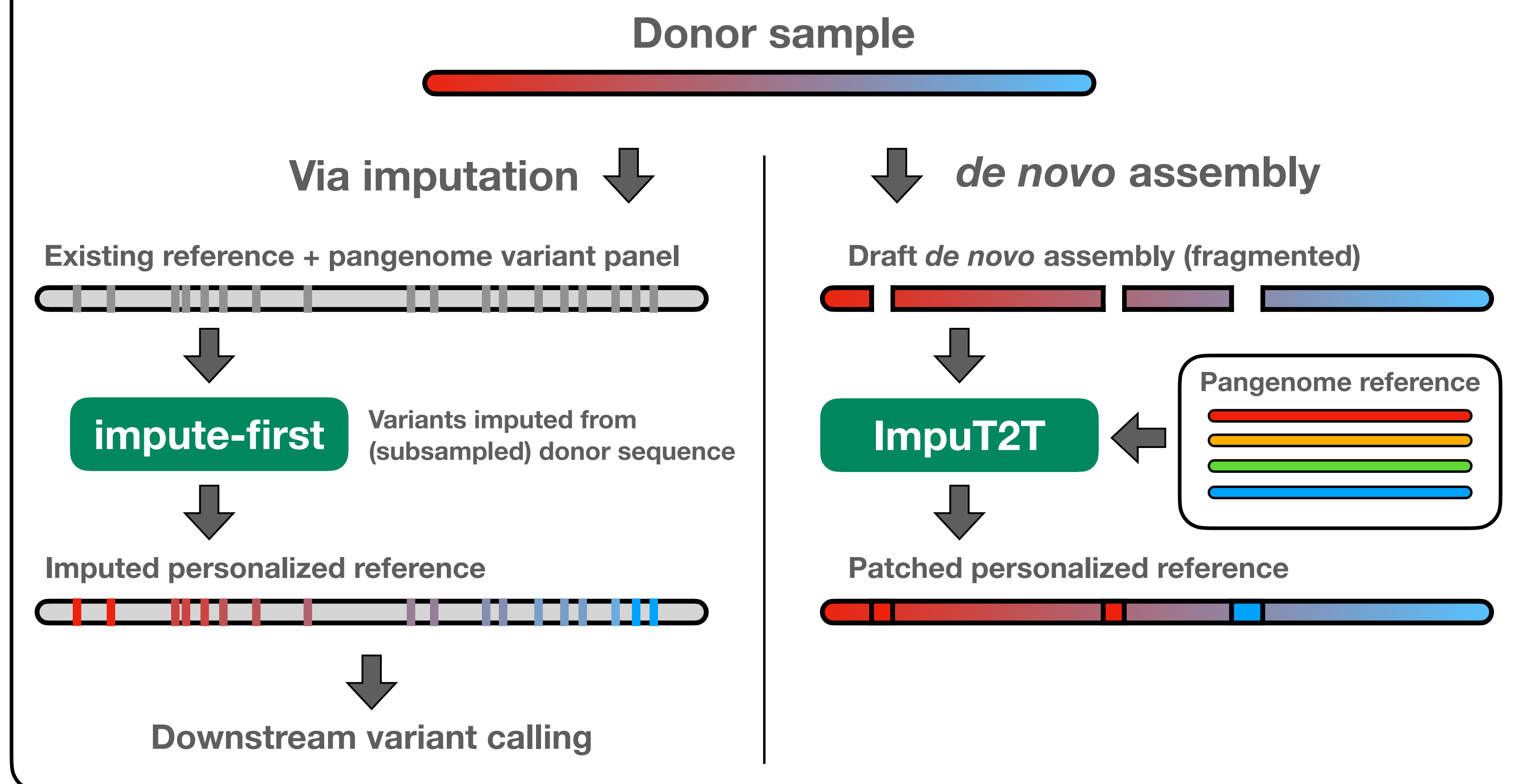
Reducing reference bias

Better Reference

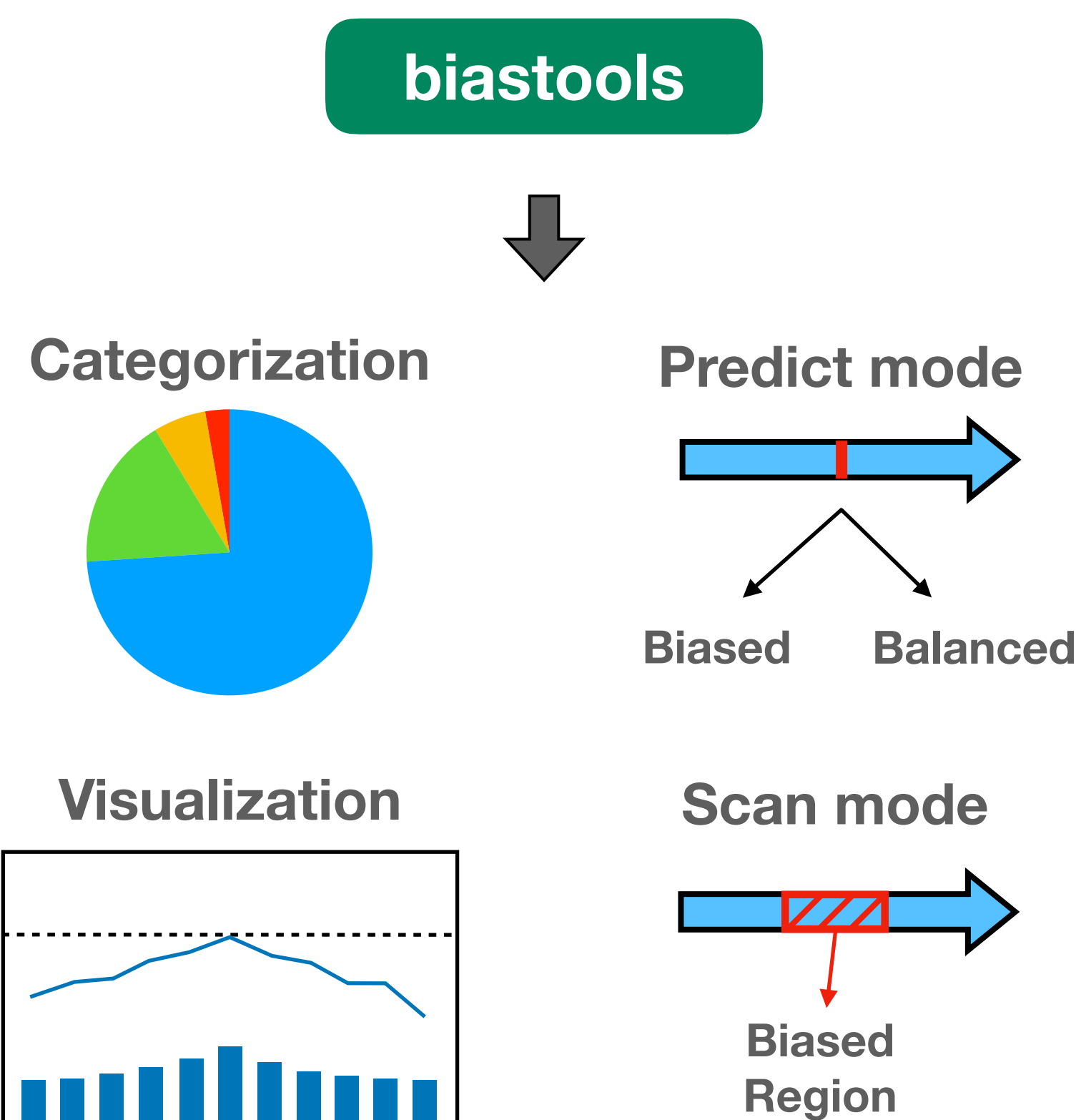
General approach

Complex loci

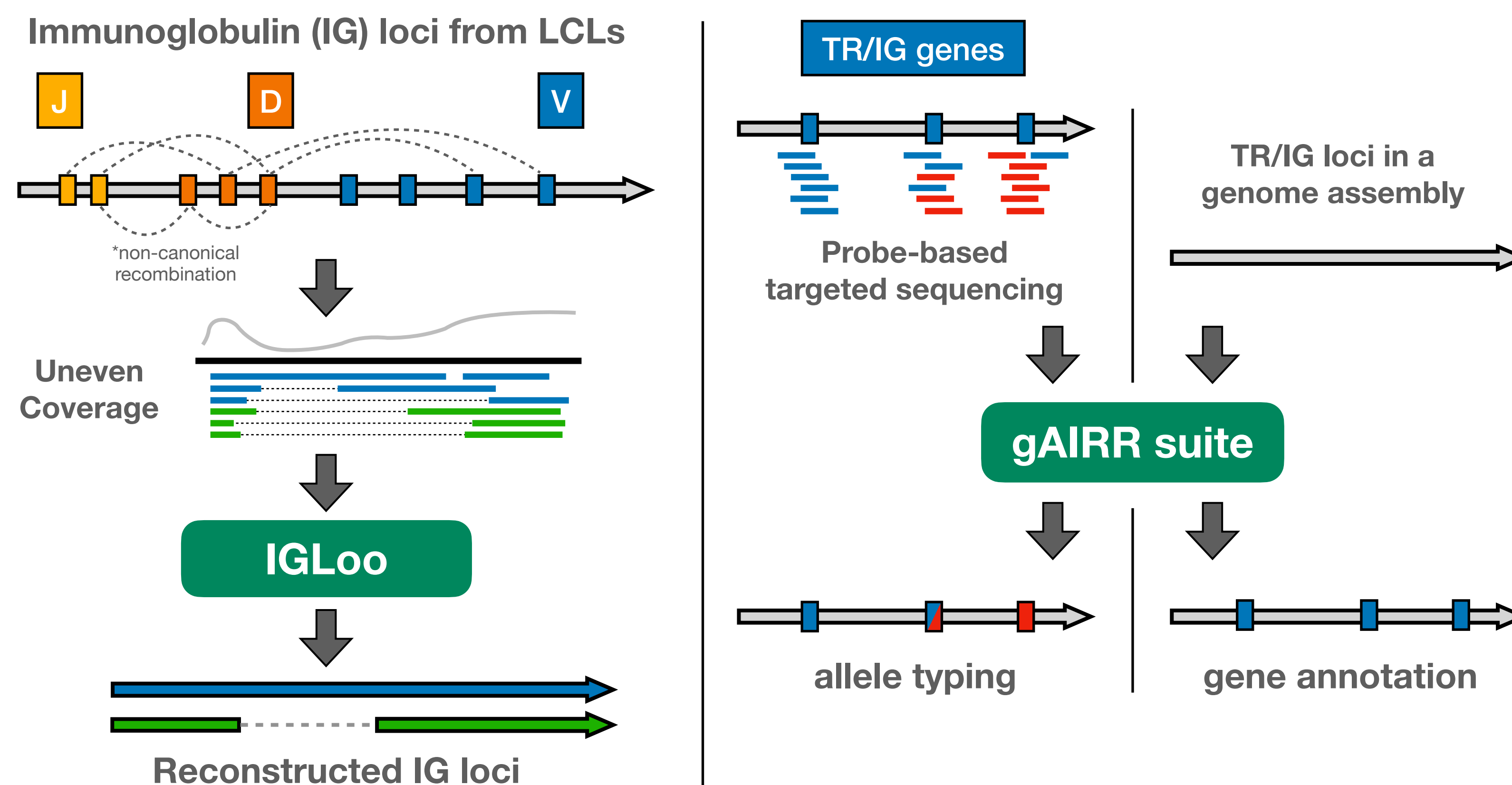
2 Build and use personalized references



1 Measuring reference bias



3 Complex immune loci



Methods developed

Conceptual connections